Deep Learning Optimized on Jean Zay

PyTorch profiler

IDRIS

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- We use a profiler to monitor an execution.
- It allows us to know the **time** and **memory** consumed by each part of the code.
- The results returned by the profiler point to the weaknesses of our code and tell us which parts we should **optimize** in priority.
- The profiler is a wrapper which records various information during the execution of the code.

This could be slowed down depending on the requested traces.
We usually monitor only **a few training steps**.

```
with prof:
        for epoch in range(0,args.epochs):
                for i, (images, labels) in enumerate(train_loader):
                        [...]
                        prof.step()
```

PyTorch profiler

```
from torch.profiler import profile, tensorboard_trace_handler, ProfilerActivity, schedule

prof =  profile(activities=[ProfilerActivity.CPU, ProfilerActivity.CUDA],       # 1
                schedule=schedule(wait=1, warmup=1, active=5, repeat=1),        # 2
                on_trace_ready=tensorboard_trace_handler(logname),              # 3
                profile_memory=True,                                            # 4
                record_shapes=False,                                            # 5
                with_stack=False,                                               # 6
                with_flops=False)                                               # 7
```

1. We monitor the activity both on CPUs and GPUs.
2. We ignore the first step (`wait=1`) and we initialize the monitoring tools on one step (`warmup=1`). We activate the monitoring on 5 steps (`active=5`) and repeat the pattern only once (`repeat=1`).
3. We store the traces in a TensorBoard format (.json).
4. We profile the memory usage.
5. We don’t record the input shapes of the operators.
6. We don’t record call stacks (information about the active subroutines).
7. We don’t request the FLOPs estimate of the tensor operations.

TP2_2: Profiler Overview

Tutorial: https://pytorch.org/tutorials/intermediate/tensorboard_profiler_tutorial.html

Configuration

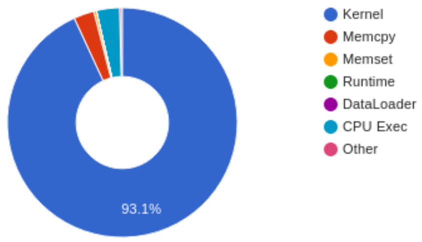
Number of Worker(s)	**1**
Device Type	**GPU**

GPU Summary ?

GPU 0:	
Name	**NVIDIA A100-SXM4-80GB**
Memory	**79.14 GB**
Compute Capability	**8.0**
GPU Utilization	**93.03 %**
Est. SM Efficiency	**92.18 %**
Est. Achieved Occupancy	**33.48 %**

Execution Summary

Category	Time Duration (us)	Percentage (%)
Average Step Time	449,011	100
Kernel	418,188	93.14
Memcpy	12,849	2.86
Memset	1,620	0.36
Runtime	0	0
DataLoader	0	0
CPU Exec	14,672	3.27
Other	1,682	0.37



Type and memory capacity of the GPU

% of time spent with an active GPU

% of active SMs

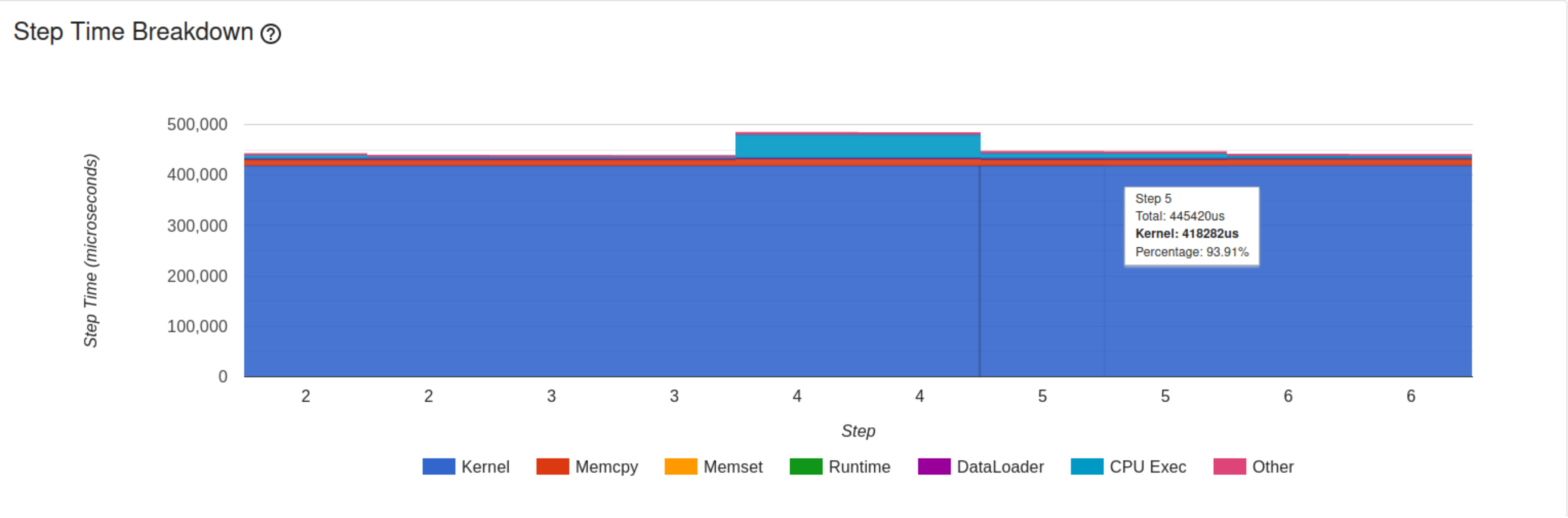
% of active wraps on an SM

A100

Streaming Multiprocessor

[Link to image]()

TP2_2: Profiler Step Time Breakdown

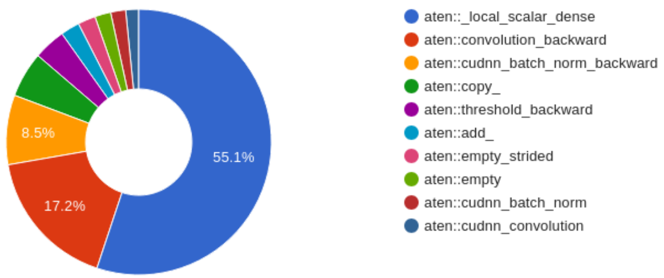


TP2_2: Profiler Operator View

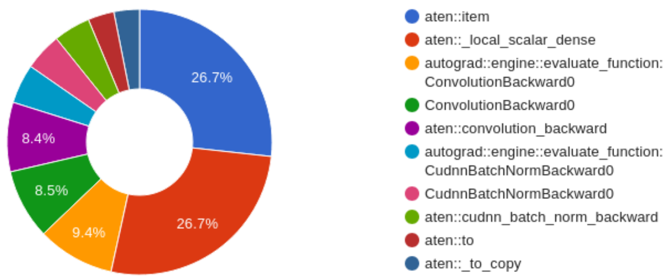
Operator View

All operators · Top operators to show 10

Host Self Time (us)



Host Total Time (us)



Group By: Operator

Search by Name

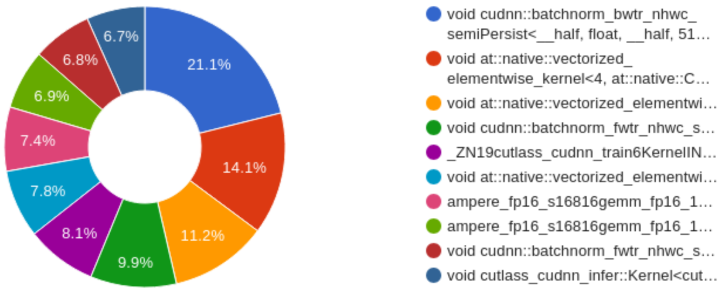
Name	Calls	Device Self Duration (us)	Device Total Duration (us)	Host Self Duration (us)	Host Total Duration (us)	Tensor Cores Eligible	Tensor Cores Self(%)	Tensor Cores Total(%)	
aten::cudnn_convolution	775	0	0	31458	31458	Yes	0	0	View CallStack
aten::_convolution	775	0	0	3146	34604	Yes	0	0	View CallStack
aten::convolution	775	0	0	4571	39175	Yes	0	0	View CallStack
aten::conv2d	1550	0	0	3895	105907	Yes	0	0	View CallStack
aten::addmm	5	0	0	265	265	Yes	0	0	View CallStack

TP2_2: Profiler Kernel View

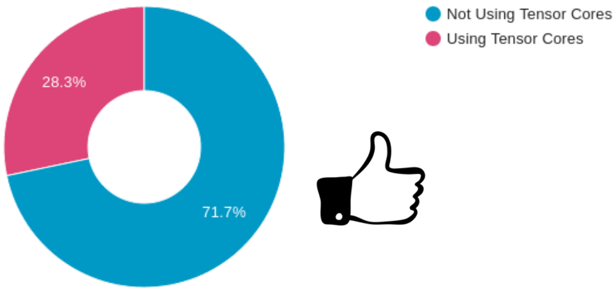
Kernel View

All kernels | Top kernels to show 10

Total Time (us)



Tensor Cores Utilization

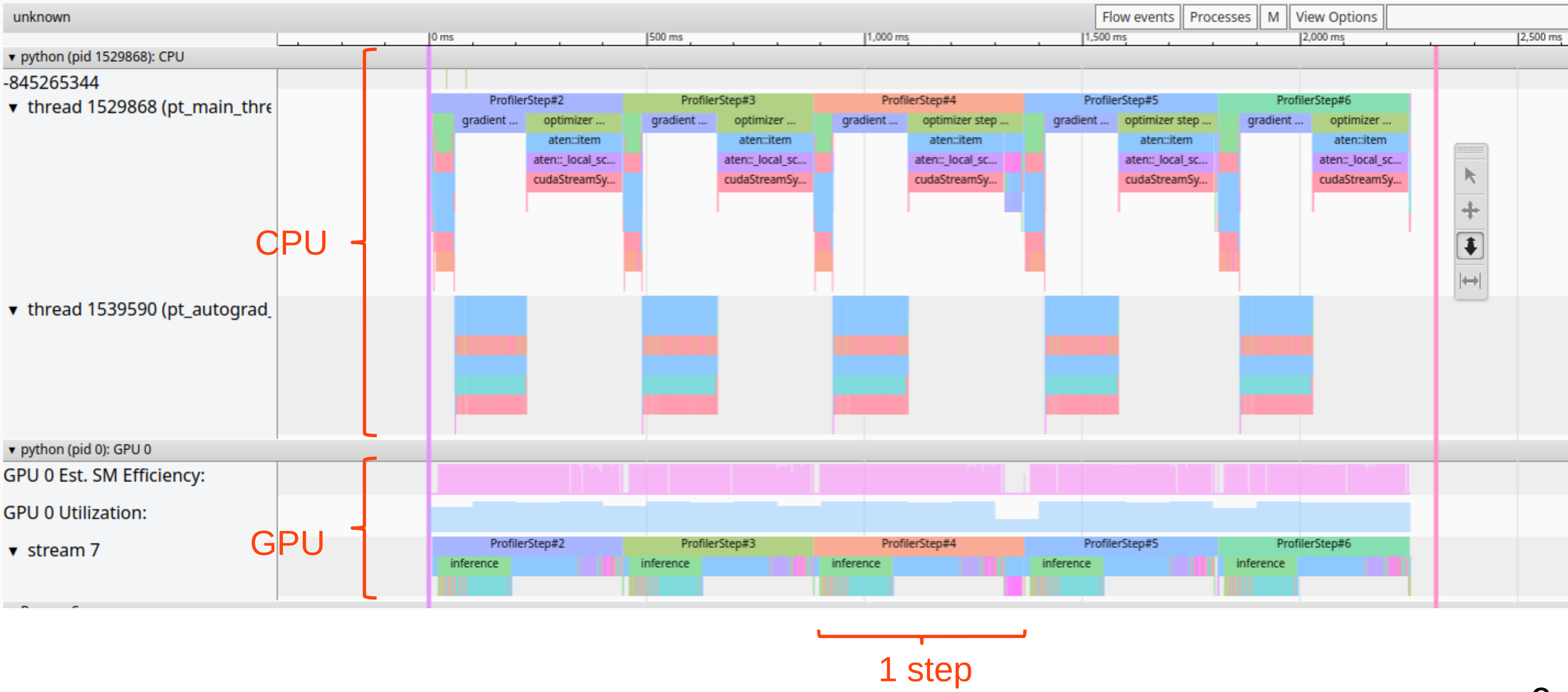


Group By
Kernel Name

Search by Kernel Name

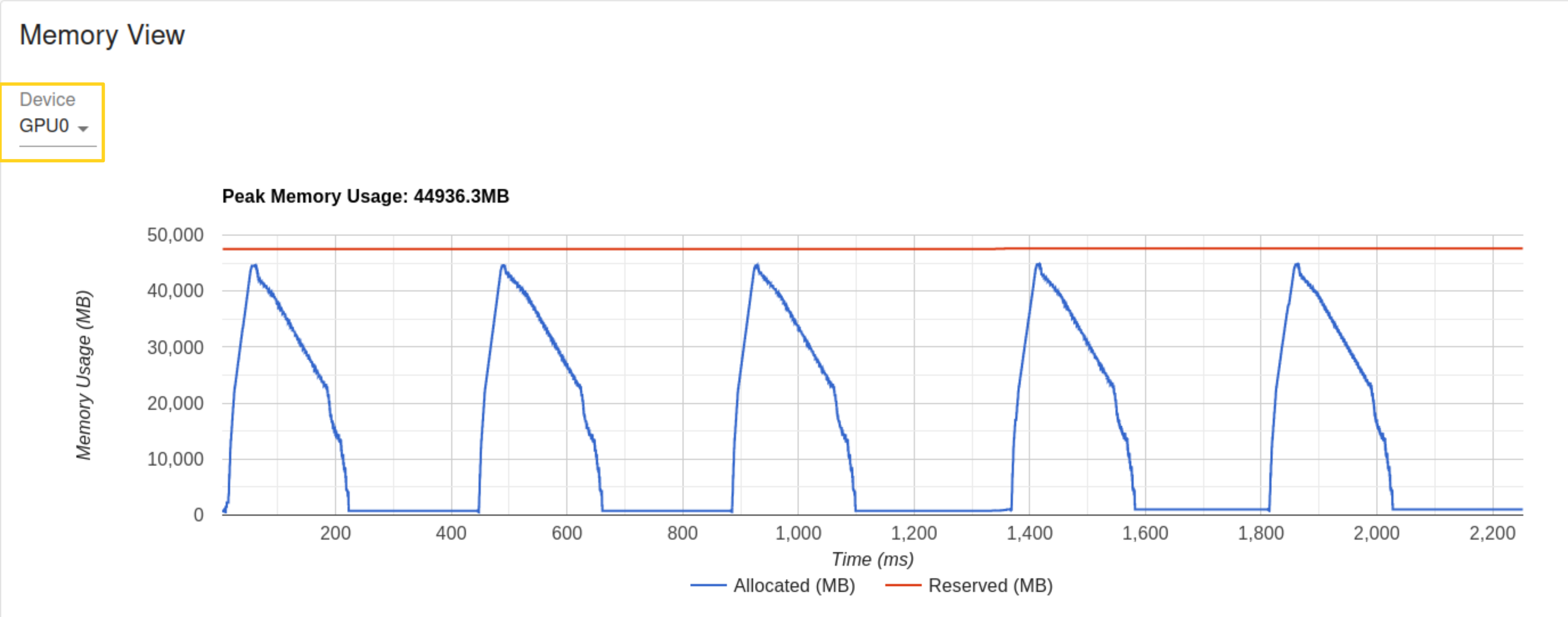
Name	Tensor Cores Used	Calls	Total Duration (us)	Mean Duration (us)	Max Duration (us)	Min Duration (us)	Mean Blocks Per SM	Mean Est. Achieved Occupancy (%)
ampere_fp16_s16816gemm_fp16_128x128_ldg8_f2f_stages_32x5_tn	Yes	430	119562	278	773	223	39.41	0
ampere_fp16_s16816gemm_fp16_128x128_ldg8_f2f_stages_32x5_nn	Yes	415	111298	268	562	230	39.14	0
void cutlass_cudnn_infer::Kernel<cutlass_tensorop_f16_s16816dgrad_optimized_f16_128x128_32x4_nhwc_unity_stride_align8>(cutlass_tensorop_f16_s16816dgrad_optimized_f16_128x128_32x4_nhwc_unity_stride_align8::Params)	Yes	230	108892	473	758	450	31.52	0

TP2_2: Profiler Trace

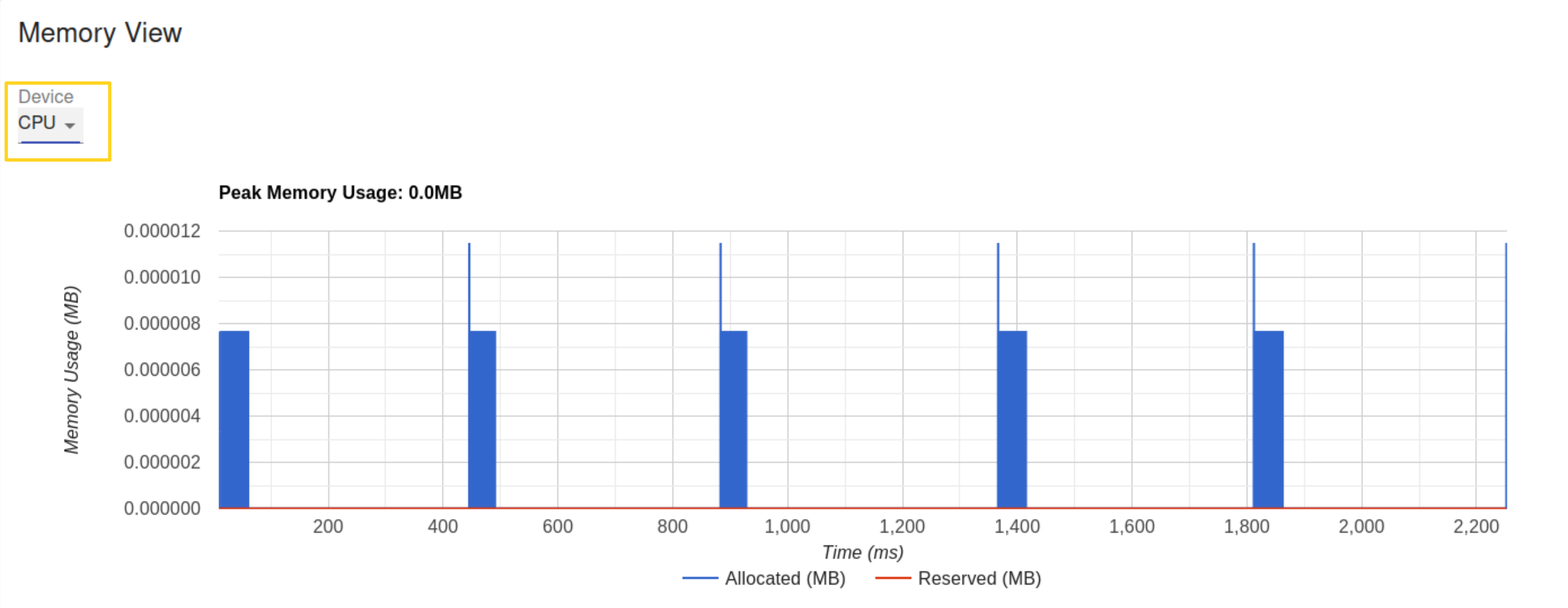


TP2_2: Profiler Trace (1 step - GPU)

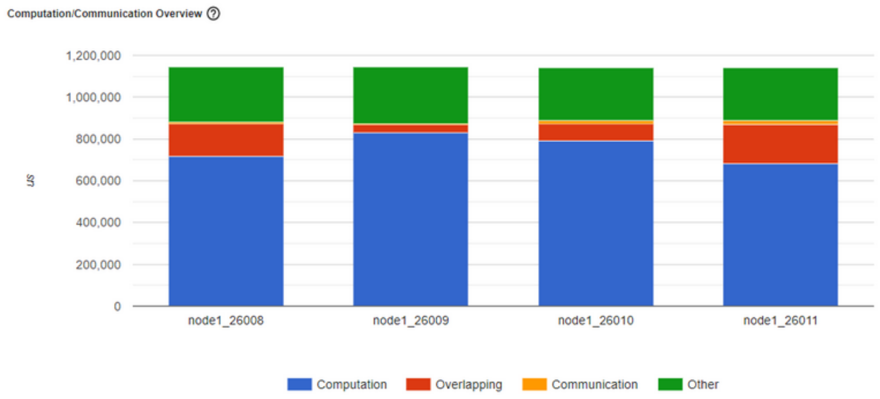
TP2_2: Profiler Memory View (GPU)



TP2_2: Profiler Memory View (CPU)



TP2_2: Profiler Distributed



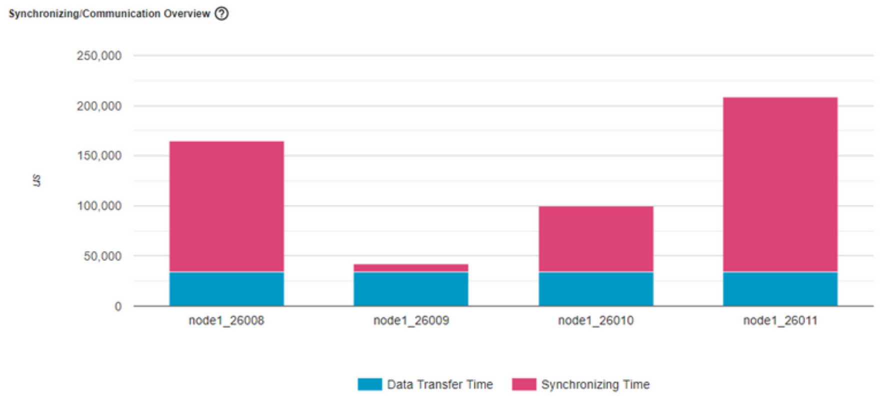


Image from the tutorial: https://pytorch.org/tutorials/intermediate/tensorboard_profiler_tutorial.html

TP2_2: PyTorch profiler

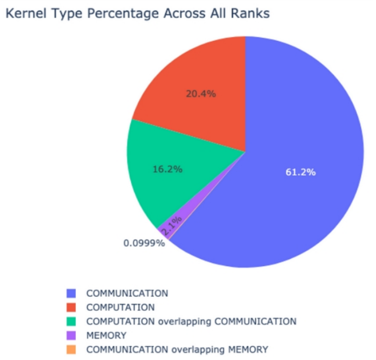
• NOTE

TensorBoard Plugin support has been deprecated, so some of these functions may not work as previously. Please take a look at the replacement, HTA.

Holistic Trace Analysis: https://hta.readthedocs.io/en/latest/

- Analyses PyTorch Profiler traces.
- Less user-friendly than TensorBoard Plugin.
- Focus on GPU usage.

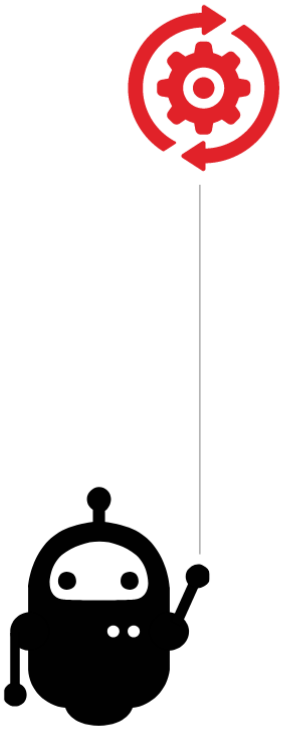
```
analyzer = TraceAnalysis(trace_dir = "/path/to/trace/folder")
kernel_type_metrics_df, kernel_metrics_df = analyzer.get_gpu_kernel_breakdown()
```




```
time_spent_df
```

	rank	idle_time(ns)	compute_time(ns)	non_compute_time(ns)	kernel_time(ns)	idle_time_pctg	compute_time_pctg	non_compute_time_pctg
0	0	552069	596651	884850	2033570	27.15	29.34	43.51
1	1	431771	596759	1004227	2032757	21.24	29.36	49.40
2	2	312107	596886	1124788	2033781	15.35	29.35	55.31
3	3	274646	604137	1154491	2033274	13.51	29.71	56.78
4	4	418833	598040	1021824	2038697	20.54	29.33	50.12
5	5	318972	601581	1112561	2033114	15.69	29.59	54.72
6	6	388040	598029	1047787	2033856	19.08	29.40	51.52
7	7	454830	599358	979022	2033210	22.37	29.48	48.15

TP2_2: PyTorch profiler



- Implement the PyTorch profiler in **`dlojz.py`**.
- Visualize the trace with TensorBoard and draw conclusions about possible optimizations.